

Internal division of a segment



1 Positive

2

a $x = \frac{mx_2 + nx_1}{m + n}$

b $y = \frac{my_2 + ny_1}{m + n}$

3 $\left(\frac{9}{4}, \frac{47}{4}\right)$

4

a $(2, 4)$

b $(0, 5)$

c $\left(\frac{1}{7}, \frac{74}{7}\right)$

5

a $\left(\frac{24}{5}, 4\right)$

b $\frac{9}{5}$

6

a $\left(\frac{14}{3}, \frac{7}{3}\right)$

b $\left(\frac{19}{3}, \frac{13}{2}\right)$

c $\frac{5}{2}$

d $\frac{5}{2}$

e Yes

External division of a segment

7 Negative

8

a $(6, 21)$

b $\left(\frac{5}{2}, -\frac{1}{2}\right)$

9 $(1, 8)$

10

a $1 : 5$

b $6 : -5$

c $-1 : 6$

11 $(2, 4)$